

LESSON GUIDE

Technology Plus

Unit 2: Programming

1.2.3 – Octal

Lesson Overview:

In this lesson, students will explore the octal number system and how it's used to manage file permissions in Linux, connecting number systems to real-world cybersecurity practices.

Students will:

- Explain the octal numbering system, including its digits, structure, and differences from decimal and binary systems.
- Describe how the octal system simplifies representations of binary data, particularly in Linux file permissions.
- Apply octal notation practically by setting and modifying Linux file permissions in a hands-on lab.

Guiding Question: How does understanding octal numbering simplify managing file permissions in computing?

Suggested Grade Levels: 8-12

Technology Needed: Yes

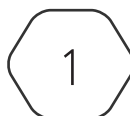
Approximate Time Required: 60-75 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 1.2 – Identify notational systems.
 - Binary
 - Hexadecimal
 - Decimal
 - Octal

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Lesson Background

Background Information:

While octal might seem like just another number system, it plays a critical role in how Linux systems manage file access and enforce permissions through access control mechanisms. File permissions aren't just a tech detail – they're a frontline defense. By unpacking octal in this context, students don't just learn math; they gain insight into how operating systems enforce rules, prevent misuse, and shape secure environments. It's a foundational concept that links low-level logic to high-stakes security.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 1.2.3 – Octal
Materials per Student	• Lab Slides 1.2.3 – Understanding Octal Linux File Permissions • Student Notes 1.2.3 – Octal • Student Handout 1.2.3 – Octal • Assessment 1.2.3 – Octal (optional)

Cyber Terms:

Term	Definition
octal	a base-8 number system using digits 0 through 7, often utilized in computing contexts such as Linux file permissions

1.2.3 – Octal

Guiding Question: How does understanding octal numbering simplify managing file permissions in computing?

Stop to Talk About Octal (15-20 minutes)

- Distribute Student Handout 1.2.3 – Octal and encourage students to complete it as you go through today's lesson.
- The lesson begins with a scenario where students find a file that has the permissions of 000. This, of course, won't make a lot of sense right now, but will be neatly addressed in the Putting It All Together portion of the lesson.
- **What is Octal?** The image on the slide is a visual cue that while octal is made of eight digits, those digits only range from 0 to 7. The slide also explains that in octal, the next place value after 7 is written as 10 (one-zero), which equals 8 in decimal.
- **Why is Octal Important?** Octal is a bit of a relic in the world of computing. It does simplify binary and is used in low-level assembly languages. Linux/UNIX file permissions is an area where you will still see octal being used.
- **How Octal Works** Like binary, decimal, and hex, octal is a positional number system. This one is based on powers of 8. While students may dismiss octal because it only uses eight digits, the place value chart on the slide shows that values escalate quickly. For a point of comparison, you could also reference the decimal place value chart to show students that powers of 8 and powers of 10 are not that far apart.
- **Converting from Octal to Decimal** and **Converting from Decimal to Octal** Students will see that the formulas here follow the patterns developed when converting decimal to either binary or hex. Moving to decimal means multiplying each digit by the value of octal place value. And moving to octal from decimal involves division and remainders.
- **Octal and Binary: A Simple Relationship** Students should find this pretty neat. The chart shows each octal digit and its binary equivalent. Just like hex, octal has a direct relationship with binary that makes it another kind of shorthand for directly expressing binary values.
- **Real-World Uses of Octal** This slide sets up students for the lab. It shows that each octal digit in a Linux/UNIX file permission represents a different group.
- The first octal digit represents the owner of the file.
- The second octal digit represents permissions granted to the file's associated group, which can include multiple users – similar to how students collaborate on projects. For example, when students create a presentation in G Suite, one student typically owns the file and then grants access to their group members. Linux handles groups similarly; a file can have an assigned group, and permissions can be set so all users in that group can access it without needing individual permissions. The sudoers group, mentioned in the lab, is just one example of how Linux groups manage user privileges.

- **Teacher Note:** The “sudoers” group name comes from the Linux command sudo—short for “super user do.” It gives certain users administrative privileges.
- The last octal digit represents permissions for everyone else on the network who might stumble across the file. Think of it like your school district's OneDrive: You can see files and might interact with them, but you didn't create them and aren't part of any group responsible for managing them.
- **Quick Trick to Remember Octal Permissions** Students typically grasp the octal concept better when they realize each permission is literally like flipping switches on or off – represented by the binary 1 or 0.
- The table on the slide shows the flipping of the switches behind the octal digit.
- If you have time, you could “flip the switches” on your board for 4, 3, and 2.
- If you do, ask students if 4, 3, or 2 seem impractical for real-life permissions.
- Students should notice that read only (4) would be pretty common, but it doesn't make a whole lot of sense to be able to write to (2) or write to and execute (3) a file that you are not allowed to read.
- 755 is a common file permission setting, but students will explore others during the lab.
- **Other Real-World Uses of Octal** Octal notation also appears in legacy computing systems, such as the PDP-8, and assembly languages like PDP-11 assembly, as well as in some system error codes—reminding us that computing builds on foundations from the past, even as it moves forward.
- The slide image is the flip switch panel of a Programmed Data Processor 11 from the 1970s.
- The PDP-11 cost \$20,000 in the 70s, but is only about as powerful as today's pocket calculator.

Lab: Understanding Octal Linux File Permissions (35-45 minutes)

- On slide 3, the orange circle doesn't include the first character in the permissions string. That character isn't part of the permission settings – it indicates the type of file. A dash (-) means it's a regular file, while a d means it's a directory (what Windows users would call a folder). You might also occasionally see other letters like l for symbolic links, but - and d are the most common.
- If you start the “wrong” machine, this lab could easily be done in Kali Linux using the exact same instructions.
- In Part 1, if students forget to include the file in their ls command, the terminal will look a little different than the image on the slide. This isn't a big deal.
- Make sure that the students type the symbolic command on slide 7 the way they see it. There should not be any spaces once they start the u=rwx,g=rwx,o=rwx portion of the command.
- It can be very common to create a file, edit the file and then use **chmod +x filename** when done. The point of the lesson is that octal allows users to set all three permissions much faster.
- On slide 11, that top line in the script they are writing (#!/bin/bash) is called a shebang and it tells the system to run the script using the Bash shell, ensuring the right interpreter handles the commands. It's not required for the script to function properly, but it is good scripting practice to include it.

- On slide 15, the option -m in the useradd command allows you to quickly create Dan as a user without having to fill out contact information. If the students forget the option -m, they will see additional prompts asking for Dan's Room Number, Phone Number, etc. They can just hit Enter without entering anything to move past those prompts.
- On slide 15, students will not see anything "happen" when entering Dan's password. As extra security, Linux doesn't use dots or asterisks when creating a password to prevent shoulder surfers from learning a password's length!
- **Part 4:** In cybersecurity, **red teams** try to find and exploit weaknesses (like overly permissive files), while **blue teams** work to defend systems and tighten things up. File permissions are a basic but powerful line of defense—and a great entry point for students to start thinking like defenders *and* attackers.
- **Putting It All Together (for the lab):** How are octal permissions preferable to symbolic permissions in Linux?

Octal permissions are faster, cleaner, and easier to use once you understand how they work. They let you set all permissions for all user groups in one short command, unlike symbolic permissions which require more typing and can be harder to read at a glance.

- What command do you use to modify file permissions?
`chmod`
- What command do you use to elevate your privilege in Linux?
`sudo`
- Which action (read, write, or execute) will this command not allow you to complete on a locked down file?
Execute. Users using sudo, or even logged in as root, cannot execute a file without execute permission.
- What is the command to lock down a file.
`chmod 000 filename`
- What is the command to allow all rights to all for a file?
`chmod 777 filename`
- What is attractive to malicious actors about files that are too permissive or locked down?

Overly permissive files (like with 777 permissions) might expose sensitive data or scripts to anyone on the system, including attackers. Locked down files (like with 000 permissions) might signal that something valuable or sensitive is being hidden, which could draw even more attention from someone looking to break in.

Putting It All Together (10 minutes)

- Wrap up by reminding students that while octal may not be as widespread as some of the other number systems studied, that doesn't mean it's not a good one to know.
- Explain the octal numbering system.

Octal is a base-8 numbering system that uses digits 0 to 7. Each digit's value depends on its position, representing increasing powers of 8.
- How does understanding octal numbering simplify managing file permissions in computing?

Understanding octal numbering simplifies file permissions because each octal digit directly corresponds to three binary bits representing read, write, and execute permissions. This makes octal a compact and clear shorthand, especially when setting Linux file permissions quickly and efficiently.
- Explain why `chmod 777 filename` gives all permissions to all users.

The **chmod 777** command assigns full permissions – read (4), write (2), and execute (1) – to the owner, group, and others. Each '7' represents the sum of these permissions (4+2+1=7), granting complete control to everyone who accesses the file.
- You find a mysterious file named `secrets.sh` on a server, and its permissions are set to 000. Explain why these permissions might raise suspicion and what risks or secrets they could imply.

Permissions set to 000 indicate extreme restriction. Such files might contain sensitive data or configurations. Malicious actors might specifically target these files, suspecting they are important. Securing files with appropriate permissions – like 755, 640, or 600 – is a safer practice.
- As a final thought, consider telling students: "Octal might not be flashy, but it's powerful – like a backstage pass to file control. When you understand those three little digits, you're not just using the system – you're managing it."